AlphaRise: A compassionate neurofeedback game for MS fatigue management

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Background

Multiple sclerosis (MS) related fatigue affects 75-90% of people with MS and is the most debilitating symptom. Current strategies show modest effectiveness. AlphaRise introduces adaptive, engaging, and compassionate neurofeedback — designed to address fatigue through real-time brain-game interaction.

Research Gap:

Current neurofeedback systems rely on fixed thresholds with minimal gamified engagement and no fatigue-aware response.

Survey (N = 146 pwMS)

MFIS -- 146 participants, 21 questions, 3 domains | alpha = 0.955

Physical fatigue
Weak muscles, needing more rest — 24.78 / 36 — 68.8%

Cognitive fatigue
Slowed thinking, difficulty concentrating — 22.60 / 40 — 56.5%

Psychosocial fatigue
Limited social activities — 4.99 / 8 — 62.4%

12% strategies very effective

96% willing to try games

R^2=.01 predictors non-significant

52.4 mean MFIS fatigue

Key Finding:

Behavioral strategies cannot predict or control MS fatigue (R^2=.01). AlphaRise engages neural mechanisms directly.

How the survey shaped AlphaRise:

- 73% want "easy to use" -- minimal UI
- 65% prefer "calming visuals" -- soft palette
- 80% prefer sessions under 10 min
- 74% already play video games

MS-informed design choices

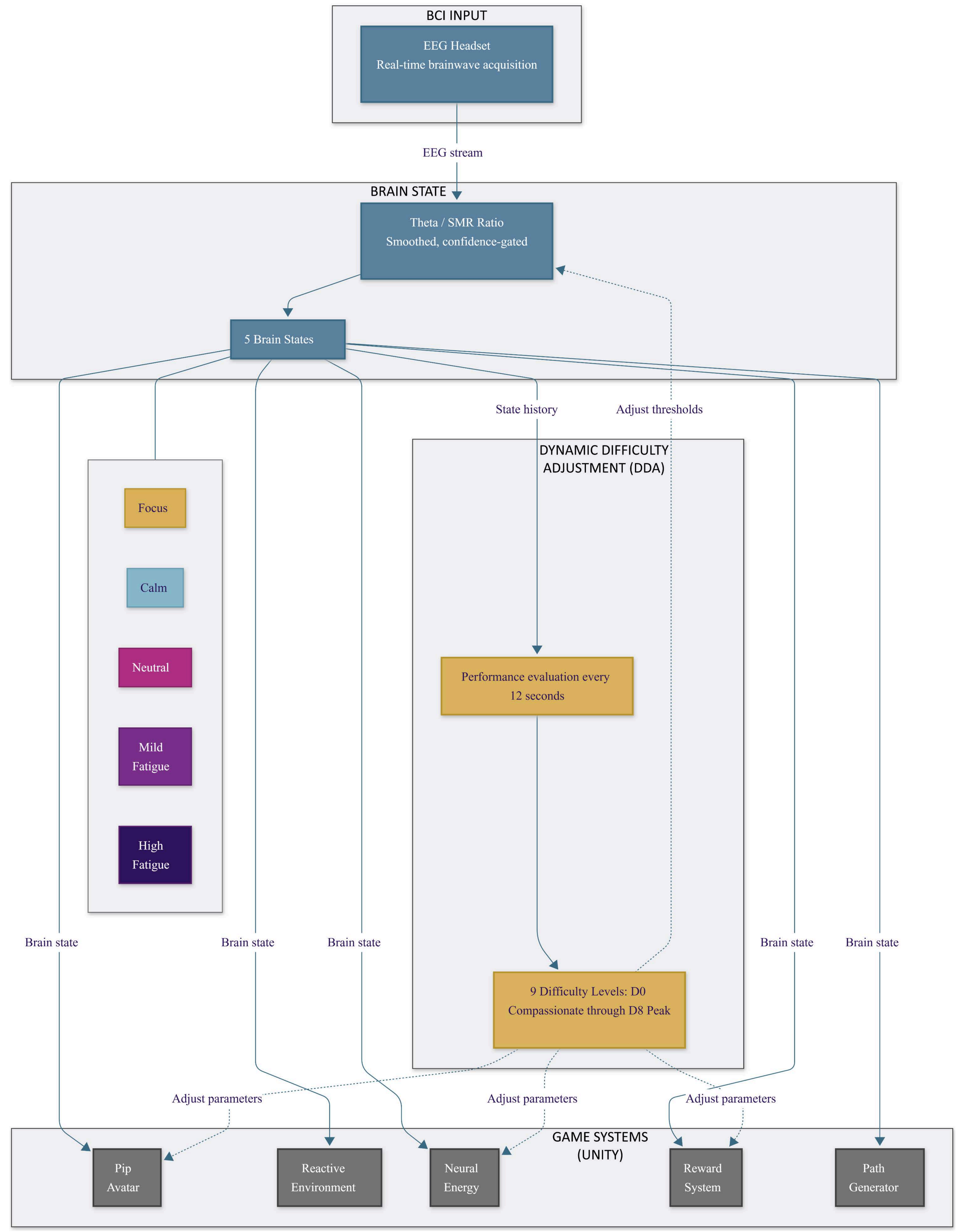
Accessible color palette
No red-green -- MS optic nerve safe

Blue-yellow dominant
High contrast for visibility
Desaturated UI, saturated game

- 5-min sessions (vs. 15-30 min prior)
- Calming 2d art, low cognitive load

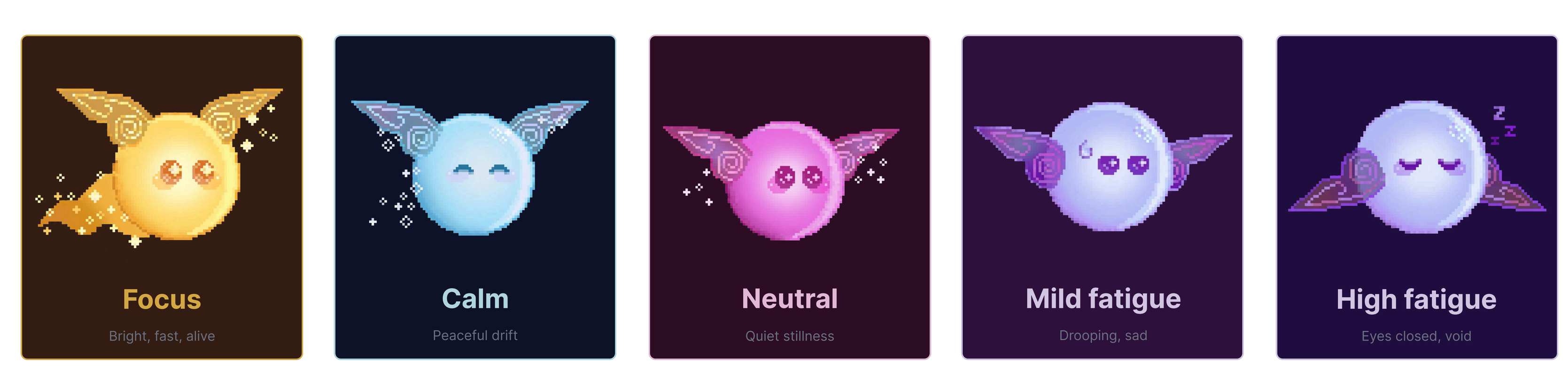
Closed-loop neurofeedback system

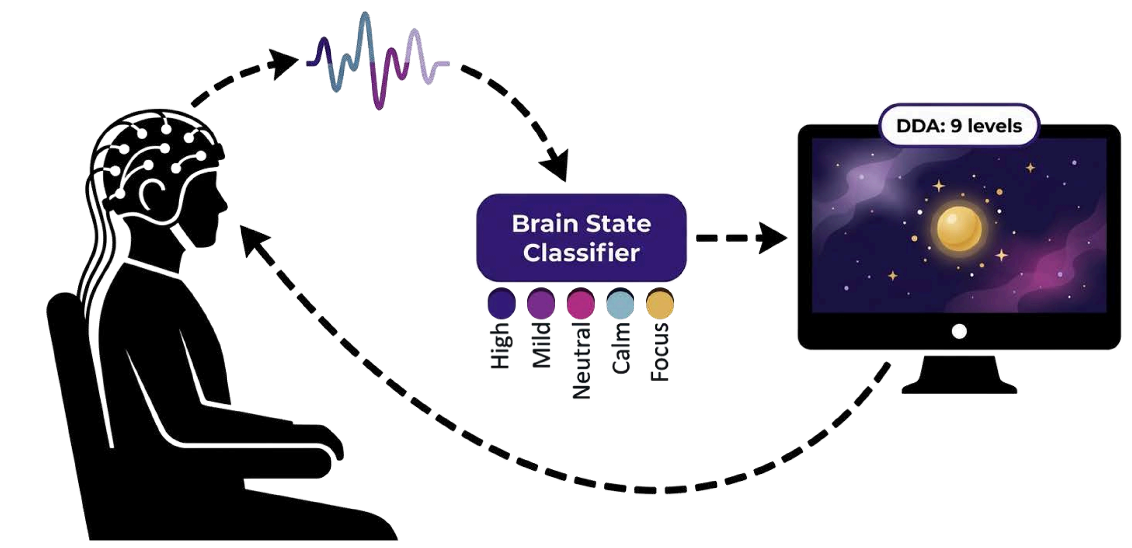
Neural activity classified into 5 states, adjusting difficulty every 12 seconds


BCI INPUT
EEG Headset
Real-time brainwave acquisition
EEG stream
BRAIN STATE
Theta / SMR Ratio
Smoothed, confidence-gated
5 Brain States
State history
Adjust thresholds
Focus
Calm
Neutral
Mild Fatigue
High Fatigue
DYNAMIC DIFFICULTY ADJUSTMENT (DDA)
Performance evaluation every 12 seconds
9 Difficulty Levels: D0 Compassionate through D8 Peak
Brain state
Brain state
Brain state
Brain state
Brain state
Adjust parameters
Adjust parameters
Adjust parameters
GAME SYSTEMS (UNITY)
Pip Avatar
Reactive Environment
Neural Energy
Reward System
Path Generator

Pip -- 5 brain states

Avatar externalizes the invisible experience of fatigue

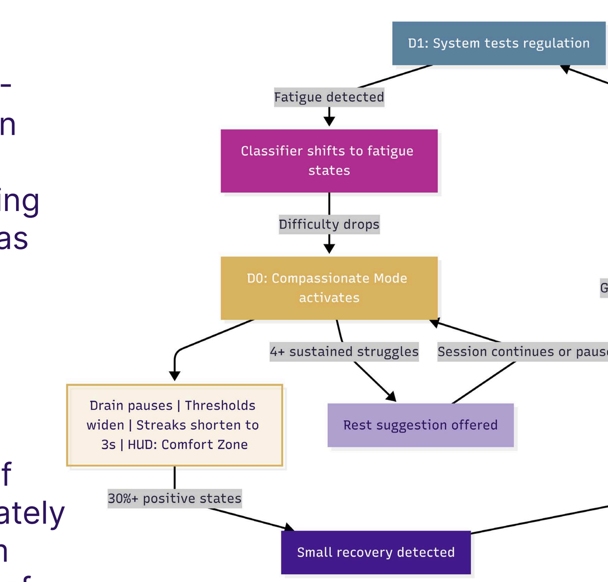

Focus
Bright, fast, alive
Calm
Peaceful drift
Neutral
Quiet stillness
Mild fatigue
Drooping, sad
High fatigue
Eyes closed, void


DDA: 9 levels
Brain State Classifier
High
Mild
Neutral
Calm
Focus

"It is okay to rest. Your brain is telling you something."

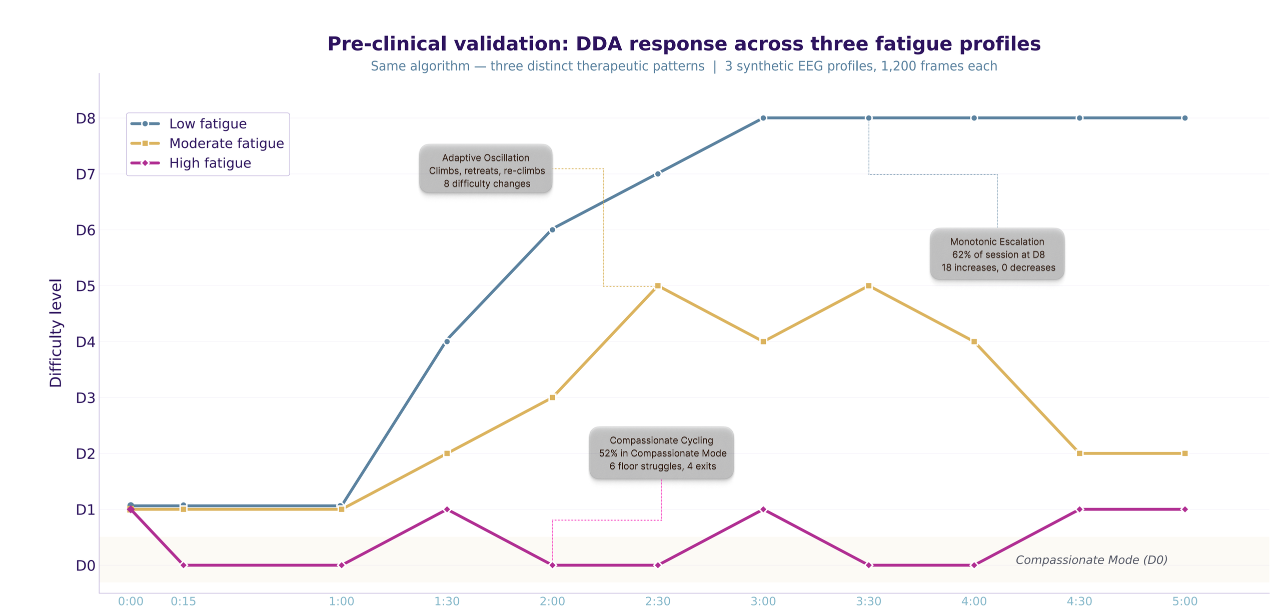
Compassionate Mode

When a player cannot sustain regulation, the game enters Compassionate Mode (D0) -- supporting rather than punishing. Energy drain pauses, classification thresholds widen, and streak requirements drop to 3 seconds, allowing even small moments of regulation to register as progress. The HUD displays "Comfort Zone" rather than any failure indicator.


D1: System tests regulation
Fatigue detected
Classifier shifts to fatigue states
Difficulty drops
D0: Compassionate Mode activates
Gentle re-test
4+ sustained struggles
Session continues or pauses
Drain pauses | Thresholds widen | Streaks shorten to 3s | HUD: Comfort Zone
Rest suggestion offered
30%+ positive states
Small recovery detected

The system continuously re-tests: when 30% positive states are detected, it returns to D1. If fatigue reasserts, it drops back to D0 immediately -- creating a gentle challenge-support rhythm that respects the energy conservation needs of MS patients.

In pre-clinical validation, the high fatigue profile spent 52% of the session in Compassionate Mode.


Pre-clinical validation: DDA response across three fatigue profiles
Same algorithm — three distinct therapeutic patterns | 3 synthetic EEG profiles, 1,200 frames each
Low fatigue
Moderate fatigue
High fatigue
Adaptive Oscillation
Climbs, retreats, re-climbs
8 difficulty changes
Monotonic Escalation
62% of session at D8
18 increases, 0 decreases
Compassionate Cycling
52% in Compassionate Mode
6 floor struggles, 4 exits
Compassionate Mode (D0)
Difficulty level
D8
D7
D6
D5
D4
D3
D2
D1
D0
0:00
0:15
1:00
1:30
2:00
2:30
3:00
3:30
4:00
4:30
5:00
Session time (5-minute session)

Comparison with prior systems

System	Adaptation	Lvls	Fatigue resp.
Birch Axon (2022)	Fixed	1	None
StoPain (2024)	Fixed	1	None
Kober MS (2019)	Clinician	Man.	None
AlphaRise (2026)	**Auto/12s**	**9**	**Compassionate Mode**

Future directions

- Clinical pilot trial with pwMS
- Live BCI integration pipeline
- Pre/post MFIS comparison
- Scalable home-based deployment

References
Scan for full reference list and acknowledgments.

vMed26
AMXRA Poster Session

Systems that care, for stories that matter.

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